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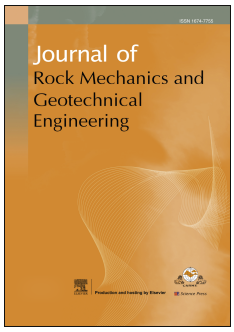

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Bayesian deep learning framework for updating landslide susceptibility assessment based on epistemic uncertainty with InSAR augmented samples

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Abstract: Data-driven modeling has emerged as an effective tool to achieve high-performance landslide susceptibility assessment (LSA). However, traditional data-driven models fail to account for the epistemic uncertainty arising from random data sampling, leading to incomplete assessments. Moreover, LSA is frequently treated as a static process, despite the evolving conditional factors and their relationships with landslide occurrence. This study introduced a novel Bayesian deep learning framework that update LSA based on epistemic uncertainty combined with InSAR techniques to achieve region-wide, completely cognizant, time-varying LSA across Zhejiang Province, China. By dividing the data into past, update, and future periods for initial model building, updating, and testing, respectively, we employed Bayesian geographically weighted convolutional neural networks (BGWCNN) to capture neighborhood effects and spatial heterogeneity while quantifying epistemic uncertainty. Recognizing the epistemic deficiencies of the initial BGWCNN, we augmented samples during the updating period in high-epistemic-entropy regions using enhanced small baseline subset InSAR. By combining augmented samples with updated landslide inventory, BGWCNN was updated (U-BGWCNN) through sequential Bayesian inference. The results indicated that BGWCNN outperformed random forest by increasing the area under the receiver operating characteristic curve (AUC) by 3.2% while ensuring the best stability. U-BGWCNN reduces 90.9% of high epistemic entropy regions and obtains the highest overall performance with an AUC of 78.8%, representing a 2.5% improvement over BGWCNN. Our results also highlight the importance of slope and NDVI for LSA in Zhejiang Province. The proposed framework ensures that LSA remains current and effective across all regions, enhancing landslide risk management and mitigation efforts.

Key words: Landslide susceptibility updating; Bayesian deep learning; Epistemic uncertainty; InSAR

1. Introduction

Landslides pose formidable threats to human lives, infrastructure, and natural ecosystems (Lin and Wang, 2018). The risk has escalated due to irrational development, vegetation degradation driven by economic growth, and extreme weather events linked to climate change (Tyagi et al., 2023; Oguz et al., 2024). Landslide susceptibility assessment (LSA) quantifies the likelihood of landslides by identifying the interrelationships between landslides and landslide conditional factors (LCFs), providing essential guidance for hazard management (Reichenbach et al., 2018). Approaches to LSA traditionally follow three principal paradigms: heuristic models based on expert judgment and empirical relations (Du et al., 2020; Nanehkaran et al., 2021); physical models grounded in the laws of physics to simulate slope stability (Ma et al., 2023; Mao et al., 2025); and recently, data-driven models that leverage computational advancements to extract insights from large datasets, demonstrating superior performance in deciphering complex data (Reichenbach et al., 2018).

Machine learning (ML) models such as logistic regression (Shao et al., 2020; Cemiloglu et al., 2023), support vector machine (Chen et al., 2016), and tree ensembles random forest (RF) (Chen et al., 2023) have prevailed in data-driven LSA with their proficiency in processing nonlinear relationships. Further, recent efforts have focused on employing deep learning (DL) models with enhanced multi-source feature extraction capabilities to improve model performance (Nanehkaran et al., 2023), for example, multilayer perceptron (MLP) (Chen et al., 2023; Nanehkaran et al., 2022), convolutional neural networks (CNNs) (Ghorbanzadeh et al., 2019), recurrent neural networks (RNNs) (Gao et al., 2024), transformer-based neural networks (Vaswani et al., 2017; Zhao et al., 2024), graph neural networks (GNNs) (Yang et al., 2024; Zhang et al., 2024), and geographic weighted neural networks (GWNNs) (Du et al., 2020). In particular, Wang et al. (2023) and Wei et al. (2024) implemented CNNs to LSA, and found that neighborhood information effectively

improved the model performance. Zhao et al. (2024) enhanced regional expression of LSA using GWNNs, highlighting the necessity of incorporating spatial importance variations into LCFs. Nevertheless, most investigated models have not synthesized these effects.

While sophisticated DL architectures have advanced the understanding of complex interrelationships, landslide risk remains inaccurately identified in areas with sparse evidence. This can be attributed to uncertainty in data sampling (Chang et al., 2023b; Huang et al., 2024), mainly caused by incomplete landslide inventory maps (LIM) from inaccessible high-elevation mountain areas (Du et al., 2020; Lin et al., 2021). The epistemic uncertainty, stemming from incomplete or imperfect knowledge, reflects confidence variations in data-driven model predictions under diverse environmental conditions (Etz et al., 2018; Hüllermeier and Waegeman, 2021). However, most data-driven models are deterministic without considering epistemic uncertainty, leading to potential miscalculations, particularly in cost-sensitive LSA (Mondini et al., 2023; Chen et al., 2024). To quantify, Zhao et al. (2023) and Chen et al. (2024) merged Bayesian theory with data-driven models by incorporating parameter uncertainty into an individual model. In parallel, Kim et al. (2018) and Di Napoli et al. (2020) employed model ensembles to quantify the epistemic uncertainty in the prediction of heterogeneous model architectures. The aggregate findings revealed deficiencies in model performance for high epistemic uncertainty regions, highlighting the value of incorporating epistemic uncertainty into LSA. Hence, mitigating epistemic uncertainty constitutes a pivotal objective for enhancing the reliability of LSA and fortifying decision-making efficacy; however, few studies have considered it.

Existing research focused on optimizing sample selection to reduce epistemic uncertainty, for instance, Sameen et al. (2020) used stratified sampling to enhance the representativeness of each LCF, and Zhang et al. (2024) applied GNN to construct balanced landslide and non-landslide samples for each environmental combination. However, the validity of these methods is unclear as the quantification of epistemic uncertainty is lacking, and no additional evidence was provided for incomplete LIM regions. Methods to supplement LIM evidence include the use of optics, synthetic aperture radar (SAR) image interpretation, or heuristic algorithms, such as hierarchical analysis (Du et al., 2020). Among these, interferometric SAR (InSAR) techniques, reflecting topographic deformation, have emerged as an effective tool for detecting landslides or dynamic slope deformations (Zhang et al., 2020; Liu et al., 2021a; Zeng et al., 2024). Meanwhile, InSAR holds advantages in large-scale identification due to its independence from cloud shading (Mondini et al., 2021; Sivasankar et al., 2021). By combining InSAR with LSA, researchers have either directly introduced the deformation velocity (DV) of InSAR to optimize the LSA results (Zhou et al., 2022; Tian et al., 2025) or supplemented the existing LIM for a more comprehensive LSA (Del Ventisette et al., 2014; Chen et al., 2023).

Given the necessity of predicting future landslide risk, the time-varying LSAs must be updated (Chang et al., 2023a; Chen et al., 2023). Traditional approaches integrate the latest time period with the original time period for modeling together (Del Ventisette et al., 2014). However, this approach can lead to ambiguous environmental conditions, as the LCFs averaged over the extended period may not reflect up-to-date states. Moreover, continuously updating and accumulating LIM necessitates repetitive modeling efforts, which is labor-intensive and time-consuming (Chang et al., 2023a). Huang et al. (2022) applied Bayesian networks (BNs) to update the prior information of past models using new evidence. In addition, transient effects and distance functions were used to incorporate the effects of historical LIM into dynamic LSA (Samia et al., 2018; Chang et al., 2023a). Nonetheless, these studies typically neglected changing LCFs and remained deficient in integrating past information, partly due to suboptimal model structure. Moreover, the studies ignored the epistemic uncertainty information embedded in the past models, which holds insights in guiding models toward selecting more valuable samples for active learning (Haut et al., 2018; Lei et al., 2022).

To remedy the shortcomings in existing studies, we propose a novel framework for dynamically updating time-varying LSA with low epistemic uncertainty using Bayesian deep learning (BDL). Specifically, we constructed a geographically weighted convolutional neural network (GWCNN) with neighborhood effects and spatial heterogeneity processing capabilities. The deterministic layers of the GWCNN were replaced with Bayesian convolutional and linear layers (BGWCNN) to quantify epistemic uncertainty. Evidence of high epistemic uncertainty regions was supplemented by InSAR, thus updating the initial model. By applying the proposed framework, researchers can continuously identify epistemic deficiencies for updated environmental conditions, thereby enabling proactive model updates based on new evidence and achieving comprehensive LSA across the study area.

2. Study area and materials

2.1. Study area

This study focused on Zhejiang Province in southeastern China (27°02′N to 31°11′N, 118°01′E to 123°10′E), bordering the East China Sea (Fig. 1). The province spans approximately 105,500 km². It features a diverse topography, including hills, mountains, fertile plains, and a complex river network.

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It receives ample precipitation, typically between 1200 and 2000 mm per year, concentrated during the plum rain season (June–July) and typhoon season (July–September). Geologically, the region exhibits a varied landscape dominated by sedimentary, igneous, and metamorphic rocks from the Paleozoic to the Neogene periods, intersected by active faults. These geographic characteristics, in conjunction with anthropogenic interventions, such as urban sprawl and resource extraction, set the stage for potential hazards. Given Zhejiang's status as an economically prosperous province with high population density, landslides pose substantial threats not only to human safety but also to infrastructure, housing, and economic sectors, leading to immense financial losses (Wang et al., 2020). Meanwhile, rapid economic development has caused major changes in environmental conditions in Zhejiang over time. Consequently, mapping and updating the landslide susceptibility map are imperative.

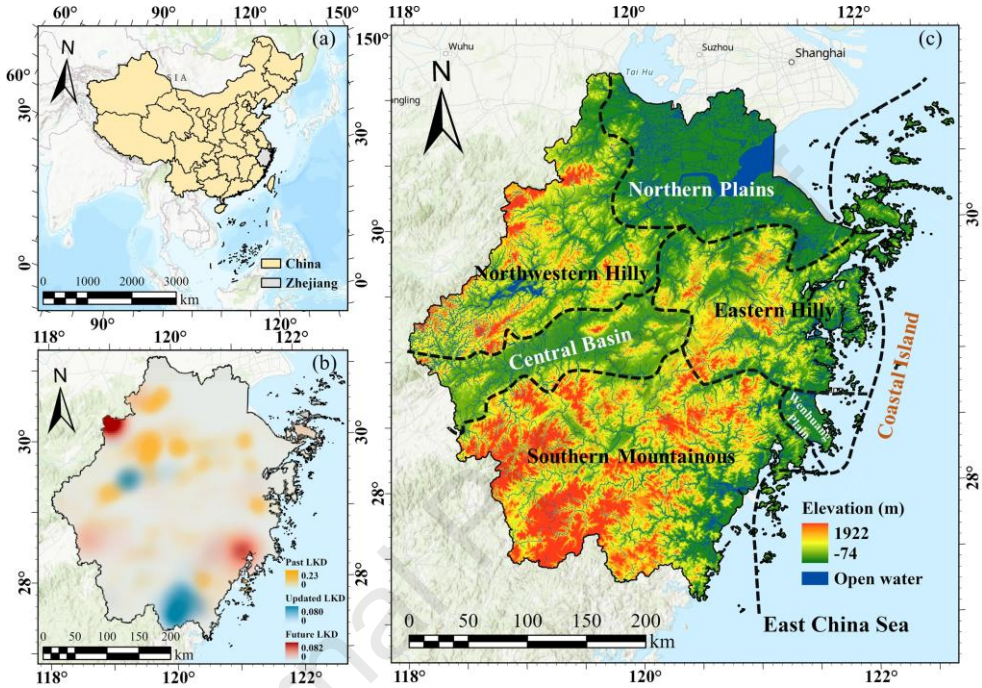

Fig. 1. Geographic location (a), kernel density distribution of LIM in different time periods (b), and major topography (c) of Zhejiang Province, China.

2.2. LIM

The LIM in Zhejiang Province is derived from historical records between 2010 and 2020, collected through field surveys. A total of 4593 landslide inventories represented by landslide centers were recorded, with 20.4% classified as collapses, 69.2% as landslides, and 10.4% as debris flows. In this study, the historical LIM was divided into three time periods: 2010–2015 (3262 landslide records), 2016–2018 (749 landslide records), and 2019–2020 (582 landslide records), representing past, present, and future periods, respectively (Fig. 1).

2.3. Landslide conditional factors

This study identified five principal categories of LCFs: topographic (elevation, aspect, slope, plan curvature, profile curvature, roughness, terrain position index (TPI), landforms), hydrologic (terrain wetness index (TWI), distance to waters (DTW), normalized difference moisture index (NDMI)), geologic (lithology and distance to faults (DTF)), ground cover (land use and land cover (LULC), normalized difference vegetation index (NDVI)), and anthropogenic influences (distance to roads (DTR)) (Figs. A1 and A2 in Appendix A). Among these, landforms, lithology, and LULC are categorized with landforms classified into peak/ridge, mountain/divide, cliff, upper/lower slopes, valleys, and further refined by sunlight exposure, slope gradients, and extent (Weiss, 2001). LULC categories encompass cropland, forests, shrubs, grasslands, water bodies, barren lands, and impervious surfaces. Lithology classification adheres to the geological map, detailing stratum, magmatic compositions, geologic structure, and lithologic character. Although the majority of LCFs are assumed to be temporally static, NDMI, NDVI, LULC, DTR, and DTW are acknowledged as dynamic, exhibiting temporal variability that reflects environmental changes in past and updated time intervals. All datasets were converted to raster data with a pixel size of 86.6 m × 86.6 m, referenced to digital elevation model (DEM) data. The data sources, types, detailed descriptions, and scales of LCFs are summarized in Table 1.

Table 1. Data source, type and resolution of landslide conditional factors.

Data type	LCF	Detailed description	Data source	Scale
Static	Elevation	The same input for all time interval	Shuttle Radar Topography Mission (SRTM-3 Version 4) DEM (https://earthexplorer.usgs.gov)	86.6 m
	Aspect			
	Slope			
	Plan curve			
	Profile curve			
	Landforms			
	Roughness			
	TPI			
	TWI			
	Lithology		Geological Survey of Zhejiang Province (https://geocloud.cgs.gov.cn/)	1:500000
	DTF			
Dynamic	LULC	Years with the most frequent landslides in past (2015) and update (2018) time interval	Annual China Land Cover Datasets (https://zenodo.org/records/8176941) (Yang and Huang, 2021)	30 m
	DTR		OpenStreetMap (https://openstreetmap.org)	1:10000
	DTW		Joint Research Centre Yearly Water Classification History (Pekel et al., 2016)	
	NDVI	Mean value during the time interval	Landsat 8 Operational Land Image Surface Reflectance data (https://earthengine.google.com)	86.6 m
	NDMI			

3. Methods

An overview of the proposed framework is shown in Fig. 2. The main steps of the framework included building an initial landslide susceptibility model using past LIM and LCFs, producing landslide susceptibility and epistemic uncertainty maps using the initial model with updated LCFs, proactively augmenting samples in epistemic deficient areas by updated evidence, and ultimately updating the initial model. The following sections explain BDL theory, InSAR-based terrain deformation analysis, and implementation details.

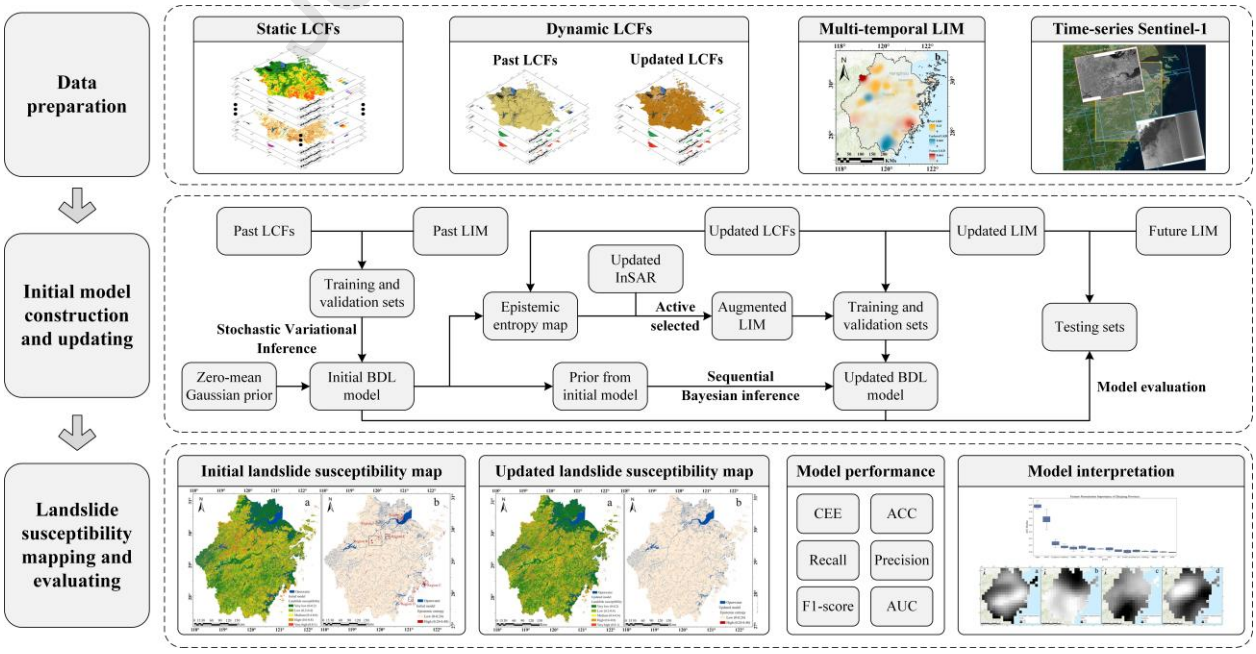

Fig. 2. Overall framework.

3.1. BDL framework

BDL offers a robust framework for incorporating Bayesian theories in DL architectures. Distinct from deterministic DL models, BDL follows two core principles: first, probability is no longer limited by the frequency of event occurrence in repeated experiments, however, rather a measure of beliefs; and second, prior beliefs influence posterior beliefs. Specifically, within the context of BDL, the Bayesian theory can be stated as

$$p(\theta|D) = p(D|\theta)p(\theta)/p(D) \quad (1)$$

where $p(\theta|D)$ denotes the posterior distribution over parameters θ given observed data D (data pairs (D_y, D_x) of labels and features); $p(D|\theta)$ is the likelihood of observing data given the model parameters; $p(\theta)$ represents the prior distribution over parameters, embodying the initial beliefs; and $p(D)$ refers to the evidence or marginal likelihood, acting as a normalization factor.

3.1.1. Functional model

The functional architecture forms the core of BDL, serving as the bridge between θ and D. Our implementation utilized GWCNN (Fig. 3), designed to simultaneously account for neighborhood information and spatial heterogeneity. The GWCNN inputs encompass both geographic localities, derived from distances relative to 260 reference points, and neighborhood LCFs within a two-pixel radius around each sample. To capture the spatial heterogeneity of LCFs, we employed a newly created geographic attention module including an MLP followed by a Sigmoid function, generating 16 geographically dependent weights corresponding one-to-one to each LCF. These weights were broadcast to align with the scale of the neighborhood LCFs, enabling element-wise multiplication. The combined results then underwent a sequence of operations including a convolutional layer, which filtered and extracted pertinent features, followed by both a max pooling and an average pooling layer, aimed at reducing dimensionality while preserving critical information. An MLP, structured with an input layer, two hidden layers, and an output layer of two neurons representing landslide and non-landslide conditions, mined more profound interrelationships among the extracted features and model output. Each convolutional and linear layer was equipped with batch normalization and activated by rectified linear units (ReLU), enhancing non-linearity and facilitating efficient training. To imbue the model with Bayesian properties, all traditional convolutional and linear layers were replaced with their Bayesian counterparts, transforming the GWCNN into a BGWCNN.

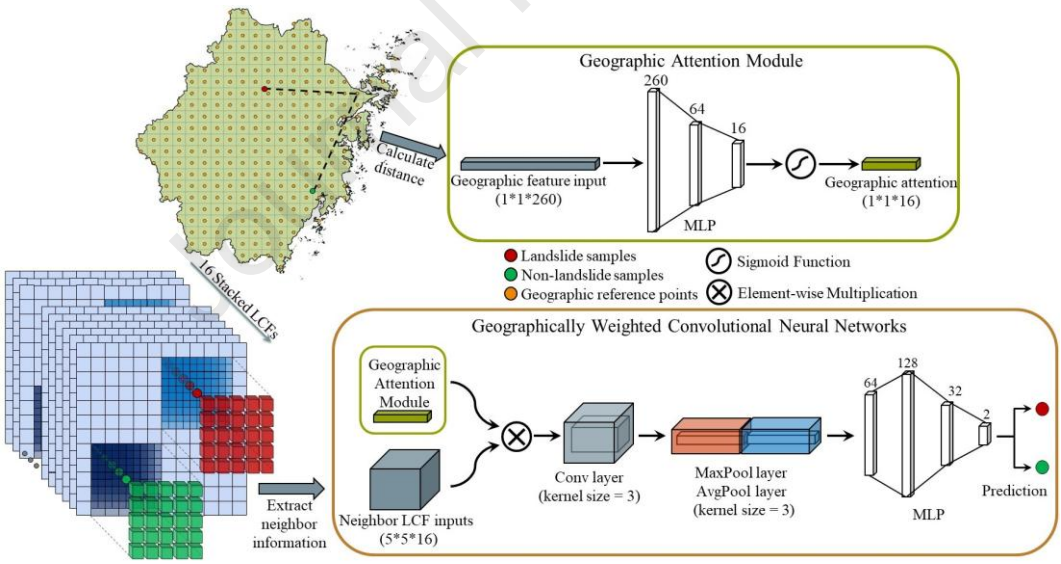

Fig. 3. Functional model structure of BDL.

3.1.2. Setting the prior for model building and updating

The prior distributions for model parameters incorporate prior knowledge and beliefs about the problem domain into BDL, functioning as a regularization mechanism and guiding the model toward plausible parameter values. For initial model construction, we typically adopt a Gaussian prior, which is equivalent to applying L2 regularization (Burden and Winkler, 2009). Specifically, zero-mean Gaussian priors with standard deviation σ were utilized:

$$p(\theta) \sim N(\theta \mid 0, \sigma^2) \quad (2)$$

As new evidence becomes available, the prior distribution evolves to reflect the latest insights. This adaptation can be achieved through sequential Bayesian inference, where the posterior from the previous iteration serves as the prior for the next. This iterative process allows the model to adapt

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dynamically to changes in the underlying data distribution and to incorporate new information efficiently:

$$p(\theta \mid D_u) \propto \int p(D_u \mid \theta) p(\theta \mid D_p)\, d\theta \quad (3)$$

where D_u and D_p denote the data available at past and update time intervals, respectively.

3.1.3. Stochastic variational inference

The specification of the prior and functional model underpins posterior calculation, yet computing the posterior often necessitates solving numerous high-dimensional integrals, which presents significant mathematical challenges and computational costs, particularly for large-scale datasets (Wang and Yeung, 2020; Jospin et al., 2022). Variational inference offers a practical alternative (Blei et al., 2017) by introducing a parameterized distribution $q(\theta;\phi)$ to approximate the true posterior $p(\theta|D)$, where θ represents the model parameters and ϕ denotes the variational parameters. Thus, the objective shifts to searching for ϕ that minimizes the Kullback-Leibler divergence (D_{KL}) between $q(\theta;\phi)$ and $p(\theta|D)$:

$$\phi = \arg\min_{\phi} D_{KL}(q(\theta;\phi)||p(\theta|D)) \quad (4)$$

Through Bayesian principles and expectation theory, this optimization problem can be reformulated as maximizing the evidence lower bound (ELBO):

$$\phi = \arg\max_{\phi}(ELBO(q) + \text{const.}) \quad (5)$$

This approach transforms the loss function in BDL to $-ELBO(q)$, encompassing D_{KL} between $q(\theta;\phi)$ and the prior $p(\theta)$, along with the expected log-likelihood. To balance these components, a hyperparameter a is introduced to control the relative importance of the D_{KL} term:

$$Loss = aD_{KL}[q(\theta;\phi)||p(\theta)] - (1-a)E_{\theta\sim q(\theta;\phi)}[\ln p(D|\theta)] \quad (6)$$

When handling large-scale datasets, traditional variational inference methods are inefficient as the whole dataset needs to be traversed to update the variational parameters. Stochastic variational inference (Stephan et al., 2017) addresses this issue by using stochastic gradient descent, which only processes mini-batches of data per iteration. During training, the reparameterization trick is employed to convert stochastic sampling into differentiable operations, thereby facilitating gradient-based optimization. The detailed derivation process can be found in Chen et al (2024).

3.1.4. Model prediction and uncertainty analysis

In BDL, the model's output for a given input is represented as a probability distribution $p(y|x,D)$ rather than a deterministic value, thereby capturing prediction uncertainty. To derive probabilistic prediction, Monte Carlo sampling is applied to the posterior distribution $(\theta|D)$, yielding multiple deterministic outputs whose average serves as the final prediction:

$$p(y|x,D) = \int p(y|x,\theta)\, p(\theta|D) d\theta \approx (1/n)\sum_1^n p(y|x,\theta_i) \quad (7)$$

The total uncertainty of BDL predictions can be decomposed into two components: aleatoric and epistemic uncertainties (Kendall and Gal, 2017). Aleatoric uncertainty stems from inherent randomness or noise within the data itself, representing uncertainty that remains even with infinite data and perfect knowledge. Epistemic uncertainty, conversely, arises from incomplete knowledge or limited training data, reflecting uncertainty about the model parameters. The appeal uncertainty can be expressed by entropy:

$$Total\ entropy = p(y|x,D)\ln p(y|x,D) \quad (8)$$

$$Aleatoric\ entropy = (1/n)\sum_1^n p(y|x,\theta_i)\ln p(y|x,\theta_i) \quad (9)$$

$$Epistemic\ entropy = Total\ entropy - Aleatoric\ entropy \quad (10)$$

3.2. Applying InSAR technology to obtain additional samples

3.2.1. Enhanced small baseline subset (SBAS) InSAR

Despite the new evidence provided by the updated LIM, the inherent randomness of landslide occurrences and incomplete LIM records can still lead to a lack of evidence in epistemic deficient areas. InSAR leverages phase difference information between SAR images acquired at different times to detect terrain changes, making it the optimal choice for rapidly locating slow-moving and unrecorded landslides that are often missing in conventional LIMs (Mondini et al., 2021; Novellino et al., 2021; Chen et al., 2023). Persistent scatters (PS) (Ferretti et al., 2001) and SBAS InSAR (Berardino et al., 2002) are the commonly used methods for processing multi-temporal SAR images and solving issues of temporal, spatial decorrelation, and atmospheric noise. Nonetheless, a single InSAR method still has shortcomings in dealing with areas of high vegetation cover and complex terrain. To address, enhanced-SBAS-InSAR (E-SBAS-InSAR) (Yang et al., 2024) integrates both PS and distributed scatters measurements, improving spatial coverage and providing more comprehensive deformation products. In the E-SBAS-InSAR processing chain, SBAS not only includes time series of terrain deformation but also estimates residual atmospheric phase delays impacting interferometric measurements after initial corrections. The process involves using the PS method on calibrated interferograms to process time displacements associated with PS points, removing contributions from topography, deformation, and residual

atmosphere estimated by SBAS.

For the application, C-band Sentinel-1 SAR satellite data operated by the European Space Agency were utilized due to their accessibility and high-resolution capabilities; given the scarcity of descending orbit data, two views of 44 ascending orbit images from Sentinel-1A with a 24-d interval during the update period from 2016 to 2018 were selected. The VV polarization C-band Sentinel-1A data were obtained from the Alaska Satellite Facility (https://search.asf.alaska.edu/), whereas precise orbital information was sourced from ESA's Sentinel-1 auxiliary data repository (http://step.esa.int/auxdata/orbits/Sentinel-1/). A 30 m resolution DEM from SRTM-1 Version 3 (https://earthexplorer.usgs.gov/) was employed for accurate geocoding and removal of topographic phase components. To mitigate atmospheric effects, the Generic Atmospheric Correction Online Service (http://www.gacos.net/) was applied to correct tropospheric delays (Yu et al., 2018a, b). During processing, multi-looking parameters were set with Azimuth Looks at 2 and Range Looks at 7, alongside a time baseline threshold of 120 d to minimize temporal decorrelation. Interferograms were generated and filtered using the Goldstein filter to reduce speckle noise while preserving phase discontinuities. Phase unwrapping was conducted using the minimum cost flow algorithm, considering only areas with coherence above a threshold of 0.3 to ensure reliable unwrapping and maintain data quality. The process was repeated twice, with the first iteration used to establish a reliable ground control point for the final process. Terrain deformation intensity was represented by line-of-sight DV over the observation period, facilitating comprehensive landslide monitoring and analysis.

3.2.2. Augmenting samples using InSAR techniques

An annual absolute DV threshold of 10 mm to 20 mm is frequently used for the preliminary identification of landslide locations (Li et al., 2021; Liu et al., 2021a; Su et al., 2022). Following this guideline, we referred to the updated LIM and manually established criteria to differentiate stable regions (urban areas characterized by low DV). We classified areas with an annual DV in the range of -12 mm to 12 mm as stable. Conversely, regions with a DV exceeding 12 mm/year or falling below -12 mm/year, coupled with slopes steeper than 5°, were identified as potential landslide zones. We categorized the remaining areas and areas lacking InSAR data (with low risk due to substantial vegetation cover) as candidate regions for non-landslide samples. By combining high-resolution time-series imagery from Google Earth and topographic maps with suspected landslide locations, we eliminated deformations resulting from other activities such as construction. Moreover, we classified suspected landslides exhibiting clear surface geomorphic features as confirmed landslides, while those in landslide-prone areas (such as exposed rocks and steep cliffs) underwent were further screened using a stricter coherence threshold of 0.5 (Li et al., 2021, 2022). The detailed sample augmentation procedure using InSAR is illustrated in Fig. 4.

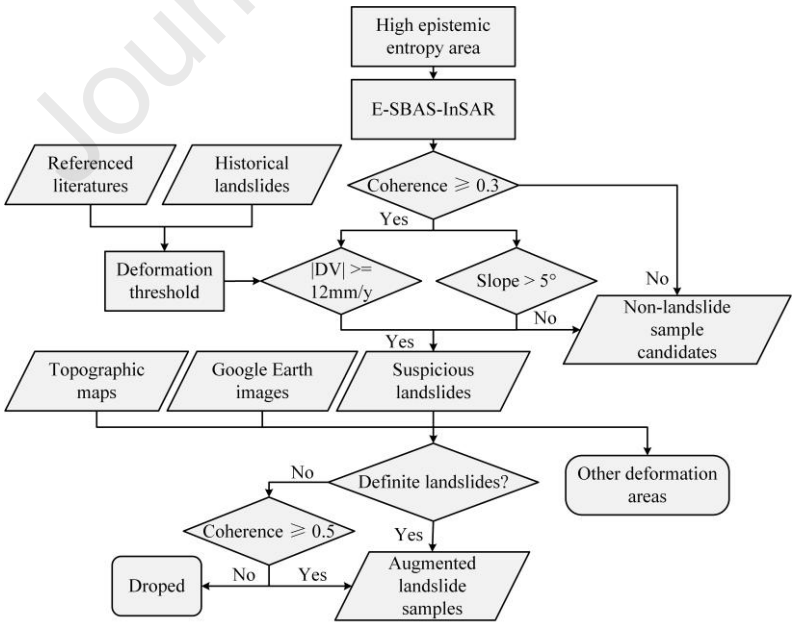

Fig. 4. Flowchart of InSAR augmented samples.

3.3. Implementation details

3.3.1. Dataset construction

The landslide susceptibility modeling in this study adopted a binary classification framework, where landslide samples were assigned the value “1”

and non-landslide samples as "0". By complementing randomly selected non-landslide samples in landslide-free areas, we constructed balanced datasets for the initial and updated models. We excluded water bodies and areas within a 500-m buffer of LIM from landslide-free areas to reduce the impact of potentially misclassified landslides and to improve model specificity. For the initial model dataset, we utilized past LIM alongside a conservative selection of non-landslide data, whereby 30% of non-landslide samples were chosen from areas with slopes less than 5°. The dataset for the updating model integrated two components: augmented landslide and non-landslide samples, combined with the updated LIM with randomly selected non-landslide samples. We divided the past datasets into training and validation sets following an 8:2 ratio. Conversely, the updated datasets were further divided into training, validation, and testing sets in the ratio of 6:2:2. The final testing set encompassed updated testing datasets, all future LIM, and randomly sampled non-landslide samples, designed to rigorously evaluate the efficacy of the model in real-world landslide prevention scenarios.

3.3.2. Data preprocessing

Considering the high tolerance of advanced DL algorithms against correlation and redundancy between LCFs, we adhere to Huang et al. (2024) without utilizing the feature selection strategy to prevent LCF information loss. Regarding categorical LCFs, we assigned frequency ratios to their respective categories, informed by the LIM of training sets (Huang et al., 2021). All LCF and geographic distance input underwent standardization via z-score normalization, with outliers (standard deviations > 3) replaced by the local average computed within a 5×5 neighboring cell matrix, ensuring data integrity.

3.3.3. Model training and prediction settings

The final outputs of BGWCNN were regarded as categorical distributions corresponding to the landslide and non-landslide labels, thus maximizing the likelihood of the observations. Hyperparameter optimization was systematically pursued during SVI training, with a consistent batch size of 360 and 120 training epochs, applied uniformly across both the initial and updated models. The learning rate of the initial model was set to 0.003, transitioning to a more conservative 0.001 during the model updating phase to ensure gradual convergence without significant overshoot. The weighting parameter a of 0.5 and a zero-mean Gaussian prior with a standard deviation of 2 was selected for the initial model parameters. During model updating, the weighting parameter a was set to 0.15, enabling the model to be more flexible in adapting to new evidence. To make the raw logistic output more comprehensible, we applied the SoftMax function, mapping the landslide occurrence predictions into a probabilistic scale ranging from 0 to 1.

3.3.4. Model evaluation and selection

To provide a comprehensive evaluation of model performance, we selected a suite of metrics including cross-entropy error (CEE), overall accuracy (OA), recall, precision, F1-score, and area under the receiver operating characteristic (ROC) curve (AUC). These metrics assess various aspects of model performance: overall uncertainty, classification accuracy, positive instance capture, reliability of positive predictions, the balance between precision and recall, and overall discrimination ability, respectively. Moreover, to address potential variability associated with random data partitioning, we conducted data splitting five times using different random seeds, evaluating the performance of each model through mean and standard deviation to quantify average performance and stability. Ultimately, the model exhibiting median performance was used for mapping and updating, ensuring robustness and reliability in our approach.

3.3.5. Experimental environment

All experiments were implemented on Python 3.8 and ArcGIS Pro 3.1. Pyro (Bingham et al., 2019), Pytorch 2.10 were used for BDL construction, and GDAL (Rouault et al., 2024) was used for geospatial computation. The process of training and inference was performed on an Intel(R) Core(TM) i9-9900X CPU @ 3.50 GHz, 64G memory, NVIDIA GeForce RTX 3090 server configuration using Cuda 12.1. E-SBAS-InSAR was processed using SARscape 5.7 of ENVI.

4. Results

4.1. Initial landslide susceptibility and epistemic uncertainty maps

In the initial landslide susceptibility and the corresponding epistemic entropy maps, we applied the equal interval method for classification. Landslide susceptibility was classified into five levels: very low (0–0.2), low (0.2–0.4), medium (0.4–0.6), high (0.6–0.8), and very high (0.8–1). Meanwhile, epistemic entropy was divided into two categories: low (0–0.24) and high (0.24–0.48). With the initial BGWCNN model trained under the optimal hyperparameters, we predicted the landslide susceptibility and epistemic entropy for all pixels in Zhejiang Province at updated time intervals (Fig. 5). The very low susceptibility occupied the largest 32.4% of the area, whereas the very high susceptibility area accounted for the smallest (6.5%). Overall, urban and agricultural areas located in the plains, such as the Central Basin, Northern Plain, and Wenhuang Plain, as well as certain coastal zones, exhibited

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very low susceptibility. Conversely, mountainous and hilly regions showed relatively higher susceptibility, with the Northwestern Hilly area presenting the highest. Meanwhile, some islands and mountains along the coast, including the northeast corner of the Coastal Islands and the mountains and islands adjacent to the Wenhuang Plain, also display very high landslide risk.

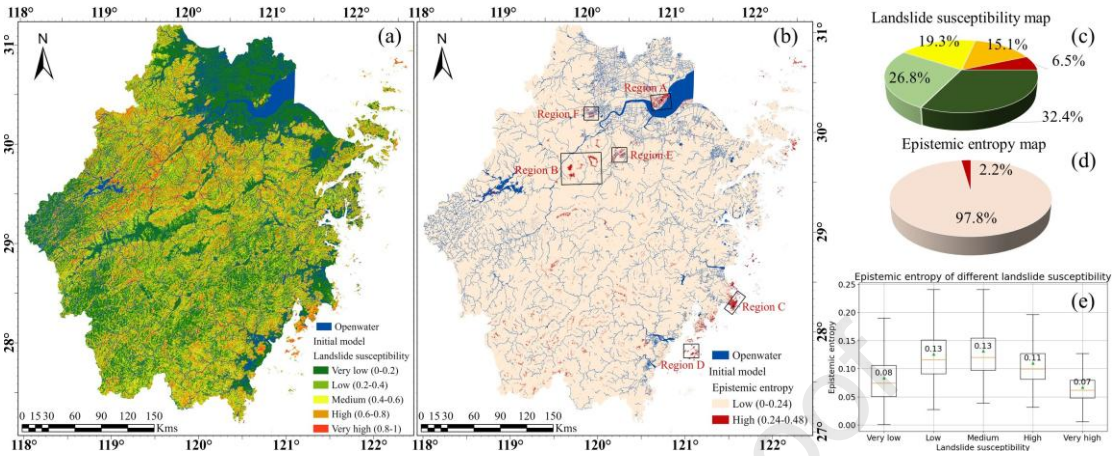

Fig. 5. Initial landslide susceptibility (a), epistemic uncertainty (b) map, and statistical links between susceptibility and entropy (c-e).

According to the epistemic entropy map (Fig. 5), the low epistemic entropy region covers 97.8% of the region, while high epistemic entropy regions account for 2.2%. Epistemic entropy peaks at medium landslide susceptibility and decreases as risk approaches either very high or very low values (Fig. 5e). High epistemic uncertainty typically occurs in mountainous regions with complex topography and areas with dense river networks around cities. In detail, hills at the estuary of the Northern Plain (Region A), the eastern part of the Northwest Hills (Region B)), and the mountains and islands of the southeastern coast (Regions C and D) exhibit the highest epistemic uncertainty. Additionally, the Baita Lake (Region E) and Xixi National Wetland Parks (Region F), with low landslide susceptibility, show high epistemic uncertainty.

4.2. Proactively augmenting samples in epistemic deficient areas

4.2.1. Ground deformation derived from InSAR techniques

To enhance the knowledge of LSA in areas with epistemic deficiencies, we used E-SBAS-InSAR to identify terrain deformation in three mountainous areas. Specifically, we obtained 151,632, 278,734, and 35,850 InSAR measurement points in regions A, B, and C, respectively (Fig. 6). By focusing on apparent deformation zones within these regions, we validated the reliability of the InSAR results against historical LIM records and engineering activities observed through Google Earth high-resolution time-series imagery (Fig. 7). Fig. 7a illustrates ground subsidence in region A caused by the conversion of paddy fields (A1) and construction processes (A2, A3). Fig. 7b shows that InSAR effectively monitored both construction activities and landslides during the observation period. Fig. 7c indicates that InSAR detected slope deformation during the update period, which later correlated with future landslide. Fig. 7d highlights extensive urban construction (A1), road and bridge construction (A2, A3), and multiple landslide occurrences (B1–B6) during the period, all associated with significant surface deformation. These findings underscore the effectiveness of InSAR for monitoring surface changes and identifying potential landslide risks, establishing the feasibility of supplementing evidence in areas with insufficient traditional LIM.

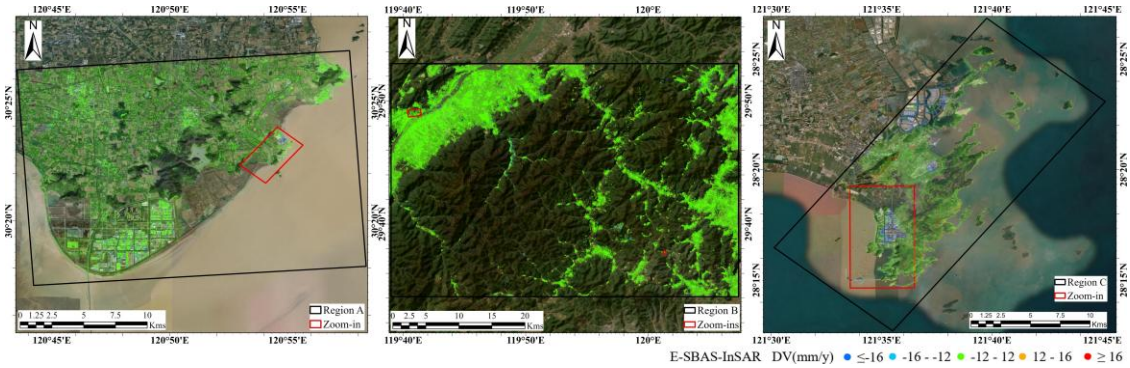

Fig. 6. InSAR deformation maps for high epistemic uncertainty areas.

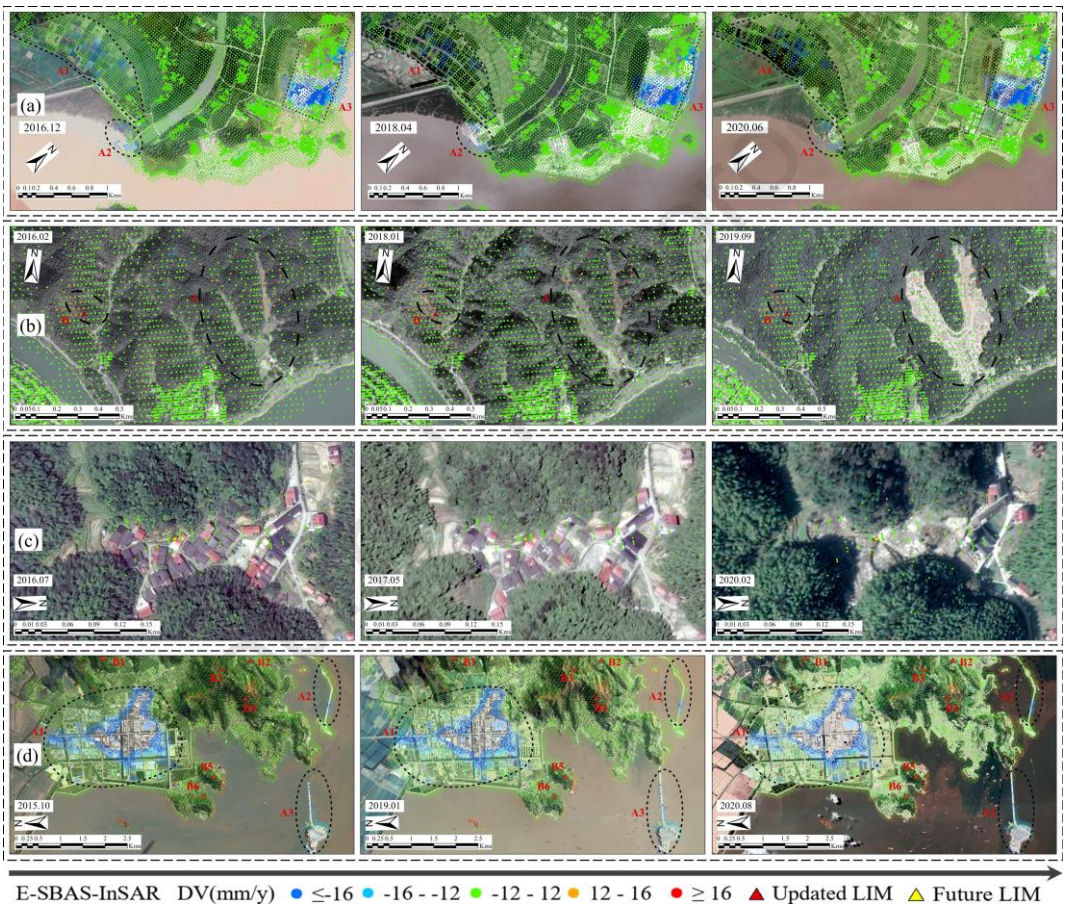

Fig. 7. Zoomed-in views of InSAR results for result validation, where (a), (b), (c) and (d) are localized zoomed-in views of regions A, B and C in Fig. 5, respectively.

4.2.2. InSAR augmented samples

By combining terrain deformation from InSAR and long-term optical images, we augmented samples in high epistemic uncertainty regions (Fig. 8). In mountainous regions exhibiting high epistemic uncertainty, zones of aggregated topographic deformation on slopes with associated landslide features were considered as augmented landslide samples (e.g. Fig. 8d and e). Several updated LIMs provided new evidence for certain areas of high epistemic uncertainty (e.g. Fig. 8f). For each region, an equal number of non-landslide samples were manually selected based on the low-risk area indicated by the initial landslide susceptibility model, and candidate non-landslide areas identified by the DV map. Moreover, given the limitations of the initial model in wetland landscapes (Regions D and E of Fig. 5b), where flat terrain poses minor landslide risk, we generated random non-landslide samples in these areas (Fig. 8b and c) to provide a more comprehensive representation. Ultimately, 191 augmented landslide samples and 208 non-landslide samples were generated within regions of high epistemic uncertainty.

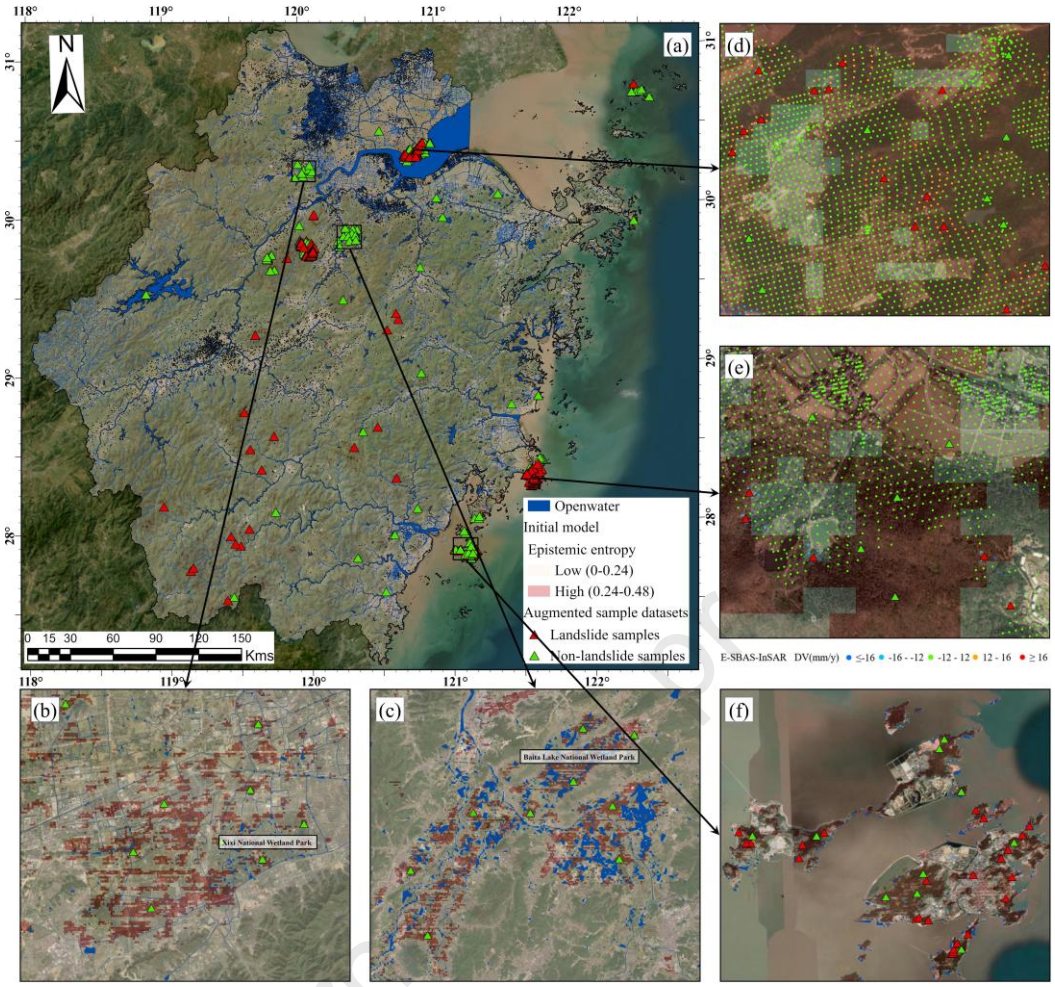

Fig. 8. Selected high epistemic uncertainty regions for sample augmentation (a) with detailed views (b-f).

4.3. Updated landslide susceptibility and epistemic uncertainty maps

By setting the same categorization thresholds as those used in the initial landslide susceptibility and epistemic entropy maps, the updated landslide susceptibility and epistemic uncertainty maps show several key improvements over the initial assessments (Fig. 9). Although the updated and initial maps were similar in distribution pattern, the updated map showed significantly higher susceptibility observed in mountainous areas compared to plains. Notably, the updated landslide susceptibility map exhibited sharper landslide occurrence probability, characterized by a 3.4% reduction in the area classified as medium susceptibility, coupled with increases of 1% and 8% in the areas categorized as very high and very low risk, respectively, compared to the initial map.

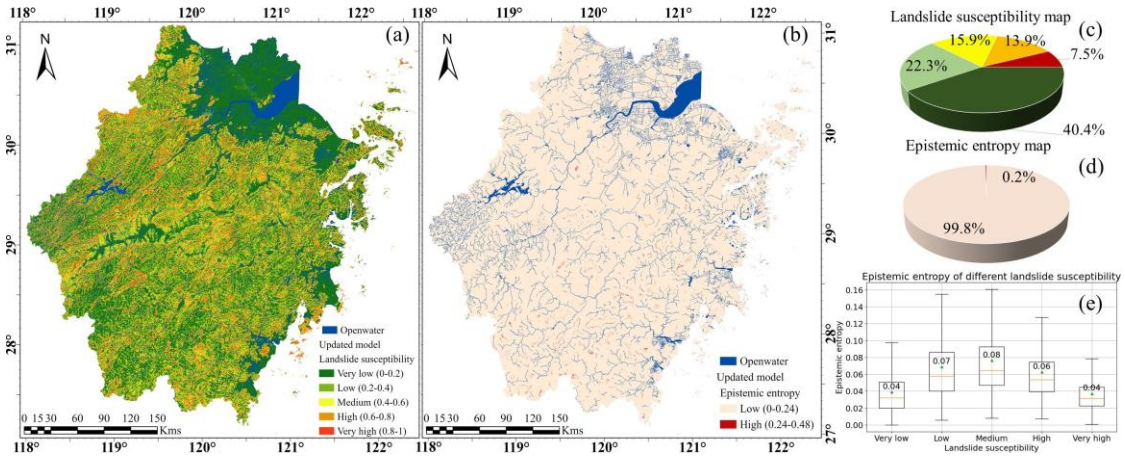

Fig. 9. Updated landslide susceptibility (a), epistemic uncertainty (b) map, and statistical links between susceptibility and entropy (c-e).

Notably, the updated epistemic uncertainty map shows marked reductions in uncertainty levels across the initial regions of high epistemic uncertainty. A comparison between the updated and initial landslide susceptibility maps in regions with high epistemic entropy revealed a significant decrease in the proportion of high epistemic entropy areas, reduced to only 0.2%, which represented 9.1% of the initially classified high epistemic entropy areas. Meanwhile, the decline in epistemic uncertainty was relatively modest for areas with very high and very low landslide risk, suggesting that these extreme categories are inherently less uncertain.

4.4. Model evaluation and comparison

To assess the effectiveness of our proposed method, we compared it with commonly used deterministic models, including RF, MLP, GWNN, CNN, GWCNN, transformer, and CNN-transformer (stacked transformers encoding modules). We also evaluated probabilistic models such as BN and GWCNN with Monte Carlo dropout (GWCNN-MC) (Gal and Ghahramani, 2016). Our results indicated that RF achieved the highest recall at 74.2%, outperforming both MLP and GWNN across all metrics (Table 2). Convolution-based neural networks including CNN, GWCNN, and CNN-transformer showed notable improvements over point-estimation-based neural networks, resulting in an AUC improvement from 3.1% to 3.2%. Although the overall performance gains from the geographic weighting module and Transformer structure were observed, these gains were less consistent with greater variability. Regarding probabilistic models, BN performed the worst among all models. BGWCNN and GWCNN-MC achieved comparable performance to GWCNN, while BGWCNN showed stronger robustness in all models. Moreover, with Bayesian updating, U-BGWCNN achieved the best CEE of 57.5%, OA of 72.4, precision of 72.9, F1 score of 71.0%, and AUC of 78.8%, which are 1.8%–4.0% higher than BGWCNN.

Table 2. Model performance comparison on testing sets.

Model	CEE (%)	OA (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)
RF	58.9±0.8	67.8±0.8	65.6±0.9	74.2±0.7	69.6±0.6	74.5±0.9
MLP	61.9±0.9	66.2±0.5	64.6±0.5	70.9±1.1	67.6±0.6	73.5±0.8
Transformer	62.3±1.3	67.1±1.4	65.5±1.6	71.8±1.9	68.5±1.2	73.7±1.1
GWNN	61.1±1.6	67.3±1.2	66.6±2.6	70.0±4.7	68.1±1.3	73.8±1.1
CNN	60.3±1.0	70.2±1.1	69.3±1.5	72.0±1.1	70.4±1.2	76.6±0.8
CNN-Transformer	60.0±1.2	70.4±1.0	72.2±2.6	65.6±2.9	68.5±1.5	77.2±0.7
GWCNN	59.6±1.2	70.8±1.1	71.8±2.9	67.8±3.3	69.6±0.9	77.0±1.1
BN	93.3±2.0	62.3±0.7	60.7±0.6	74.0±0.8	66.2±0.5	67.2±0.8
GWCNN-MC	59.7±1.1	70.3±1.3	69.8±1.4	68.8±1.6	69.1±0.9	76.7±0.7
BGWCNN	59.9±0.5	70.5±0.5	70.6±0.5	68.4±1.0	69.4±0.7	76.9±0.4
U-BGWCNN	57.5±0.6	72.4±0.7	72.9±0.8	69.6±1.2	71.0±1.0	78.8±0.8

The ROC curves offer an intuitive portrayal of model performance across various threshold settings, providing deep insights into the trade-offs

between true positive and false positive rates. As illustrated in Fig. 10, the ROC curves for GWCNN, BGWCNN, GWCNN-MC, CNN-transformer, and U-BGWCNN were closest to the upper left corner, indicating higher AUC values. However, GWCNN exhibited significant uncertainty, with AUC confidence intervals ranging from 0.759 to 0.781. In contrast, both BGWCNN and U-BGWCNN yielded relatively stable curves, with U-BGWCNN attaining the highest AUC value of 0.788.

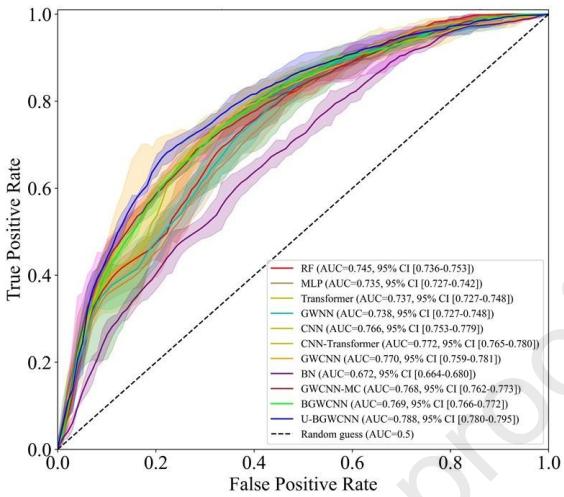

Fig. 10. ROC curves of various landslide susceptibility models. The solid line represents the average ROC by feeding five random divisions of data to each model, and the dashed line represents the 95% confidence region.

5. Discussion

5.1. Active model updating in a BDL framework

The strengths of this study lie in its seamless integration of utilizing BDL for updating LSA into a cohesive process. To handle the problem of traditional Bayesian methods not being effective in dealing with complex LSA, BDL ensures fundamental performance by leveraging DL architectures with multiple modules as the backbone. Specifically, DL models containing convolutional modules by fusing neighborhood information outperformed the traditional state-of-the-art ML model RF, which corroborates earlier studies (Wei et al., 2024; Zhao et al., 2024). The transformer-based model did not bring as much improvement to LSA as it did in natural language processing and computer vision (Vaswani et al., 2017; Wolf et al., 2020); the limited data size and lack of context may be the main reasons. While the geographically weighted module enhances performance by capturing local specificity, it exhibits sensitivity to feature noise and randomness in data partitioning. Conversely, BDL addresses this limitation by introducing parameter uncertainty, ensuring stable outputs (Chen et al., 2024; Sreeraj et al., 2024). BGWCNN demonstrates superiority over traditional CNNs primarily in its ability to handle spatial heterogeneity, model stability, and uncertainty quantification. As Bayesian layers increase the computational cost, we counted the trainable parameters, and training and inference time of all DL models (Fig. A3 in Appendix A). The incorporation of the Bayesian layer in BGWCNN resulted in a doubling of the trainable parameters compared to GWCNN, leading to increases of 1.66 s in training time and 199 s in inference time per epoch, respectively. For lightweight tasks, such as LSA with a small sample size and model parameters, the increase in computational cost is considered acceptable. Additionally, the time required for sample acquisition using BGWCNN can be reduced, as the epistemic entropy derived from BDL offers valuable guidance on areas that require sample augmentation.

The InSAR technique was used to identify slow-moving and unrecorded landslides, revealing the landslide risk in areas lacking sufficient epistemic understanding. Compared with traditional optical remote sensing image recognition and in-situ landslide inventory, InSAR with DV enables the rapid identification of suspected landslide locations. By integrating SBAS and PS techniques, E-SBAS-InSAR obtains more deformation points and achieves more accurate landslide deformation estimation (Yang et al., 2024). The E-SBAS-InSAR process facilitates a more precise and efficient initial screening of potential landslide hazards. For areas with limited historical LIM, BDL quantifies uncertainty to determine where the model holds reliability, which is essential for effective LSA (Fig. 5). The landslide identified by E-SBAS-InSAR in the area fills the knowledge gap in the region with limited data, thereby enhancing prediction reliability throughout the region. Our results indicate that updating model with InSAR-augmented samples significantly enhances

comprehension in epistemic deficient regions while boosting model performance. Similarly, InSAR technology has proven effective for both augmenting LIM and landslide early warning, offering a robust approach to identifying and monitoring landslide-prone areas (Chen et al., 2023; Wang et al., 2023). To analyze the trade-off between the InSAR augmented samples and incremental performance, we implemented a sensitivity analysis by filtering the augmented samples in ascending order based on epistemic entropy (Fig. A4 in Appendix A). The high epistemic entropy area consistently decreased as the number of InSAR samples increased, while AUC showed a decreasing trend after a continuous increase. We suggest that the potential performance degradation may be attributed to the relative reduction of samples trained in the low epistemic entropy region. Consequently, an appropriate balance of samples from high and low epistemic entropy areas must be maintained during model updating.

To further elucidate the performance improvements, we compared the performance of BGWCNN, U-BGWCNN, and U-BGWCNN* (denoting BGWCNN was updated without actively augmented samples) within high and low initial epistemic entropy areas (Table 3). Overall, all models demonstrated greater performance and stability in areas of low epistemic entropy compared to those with high epistemic entropy. Notably, BGWCNN underperformed compare to its updated counterparts, underscoring the importance of adapting models to evolving environmental conditions. Contrary to traditional LSA studies that often emphasize long-term average risks, our methodology prioritized current conditions. This focus makes it particularly advantageous for dynamic environments, where LCFs such as LULC and climate are subject to change. However, this approach may be less adept at identifying and attributing landslide risks over historical or long-term time scales. Both BGWCNN and U-BGWCNN* suffered more performance degradation in high epistemic entropy regions compared to low ones, and so did U-BGWCNN; however, the differences were much narrower. U-BGWCNN outperformed both U-BGWCNN* and BGWCNN in high epistemic entropy areas across all metrics, reinforcing the value of augmented samples in enhancing accuracy. Figs. 11 and 12 provide the overall and local zoom-in mapping results for BGWCNN and U-BGWCNN in the initial high epistemic entropy regions, respectively. U-BGWCNN effectively reduces epistemic entropy and enhances the predictability of landslide susceptibility across all examined subregions. For instance, in Regions A and B, where BGWCNN predicted numerous medium risks, U-BGWCNN provided more definitive predictions. In the zoomed-in figure, the updating samples fill the epistemic gaps of the initial model and correct the misclassification of the testing landslide and non-landslide samples in high epistemic entropy areas. While BGWCNN inaccurately classified regions C and D as uniformly high and very high risk for landslides, U-BGWCNN refined the distinctions between landslide and non-landslide samples. The detailed images show that the updated model accurately identifies non-landslide samples as low and very low risk while maintaining accurate predictions of landslide samples. Furthermore, in flatlands, such as Regions A and E, where no landslide risk exists, U-BGWCNN accurately identified these areas as very low risk, correcting the misleading low-to-moderate LSA predicted by BGWCNN. Although the enhancements in model performance between the updated and initial models were relatively modest, the practical significance of these improvements may outweigh their statistical significance. As indicated in Table 3, the majority of the improvements were observed in regions with high epistemic entropy, where U-BGWCNN achieved a 4.7% higher AUC and a 5.8% increase in OA compared to BGWCNN. Combined with visual analysis, our method effectively reduces the misclassified samples in the areas with reduced epistemic entropy of the initial model, contributing to practical landslide prevention.

Table 3. Comparison of model performance across regions with different levels of initial epistemic uncertainty. U-BGWCNN* denotes that the model was updated without actively augmenting samples.

Model	Epistemic entropy	CEE (%)	OA (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)
BGWCNN	Low	59.2±0.2	70.6±0.2	71.2±0.4	67.7±0.3	69.7±0.2	77.1±0.3
U-BGWCNN		57.3±0.2	72.5±0.4	73.5±0.4	70.6±0.5	71.4±0.3	79.2±0.2
U-BGWCNN*		57.8±0.3	72.2±0.6	72.1±0.3	74.3±0.5	72.4±0.5	78.6±0.6
BGWCNN	High	62.7±0.8	68.2±1.7	56.5±2.5	69.4±1.7	60.9±1.6	74.3±1.2
U-BGWCNN		57.9±1.2	72.2±1.0	70.4±2.8	67.1±1.4	68.3±1.5	77.8±0.8
U-BGWCNN*		61.7±1.3	68.5±1.2	68.4±3.3	66.2±1.3	67.0±1.6	74.9±1.1

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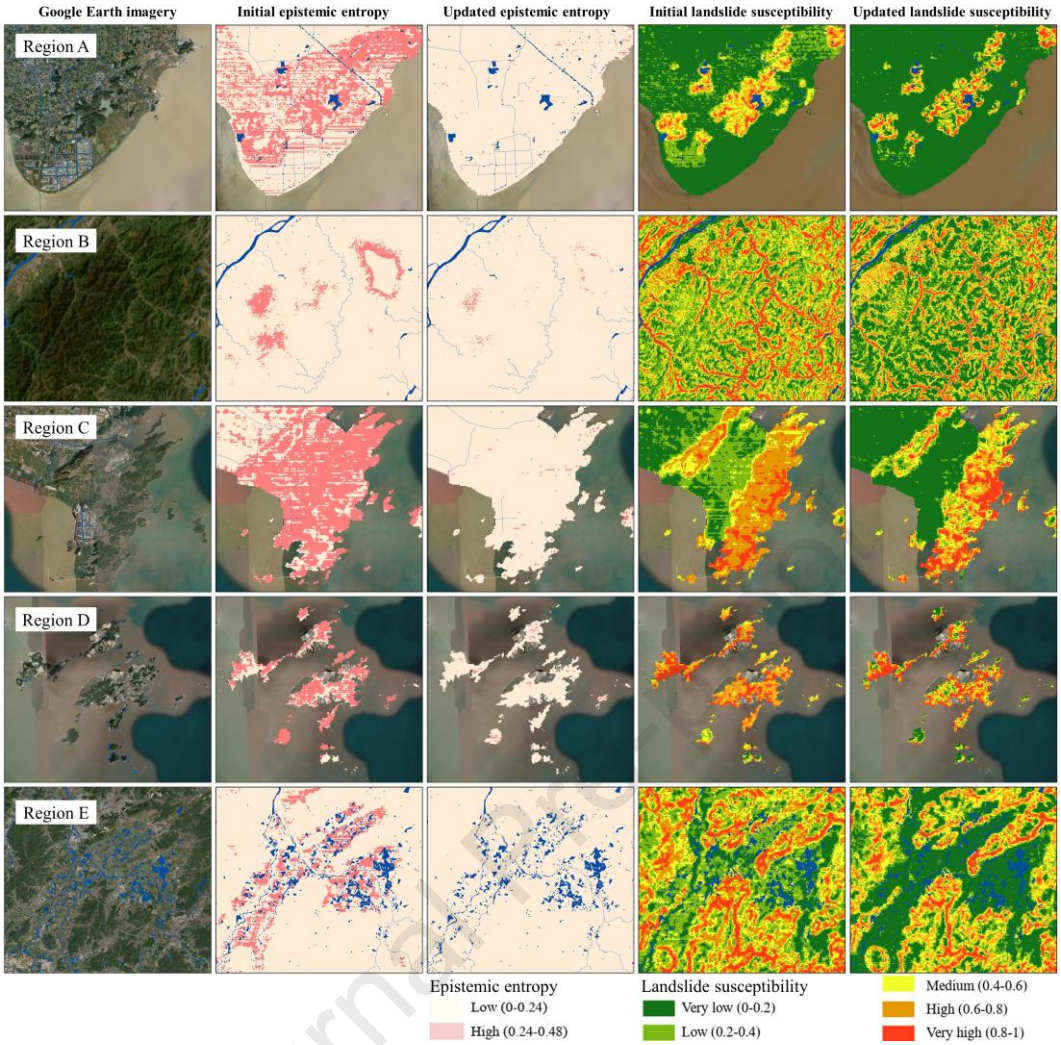

Fig. 11. Initial and updated landslide susceptibility and epistemic uncertainty maps for high epistemic entropy regions.

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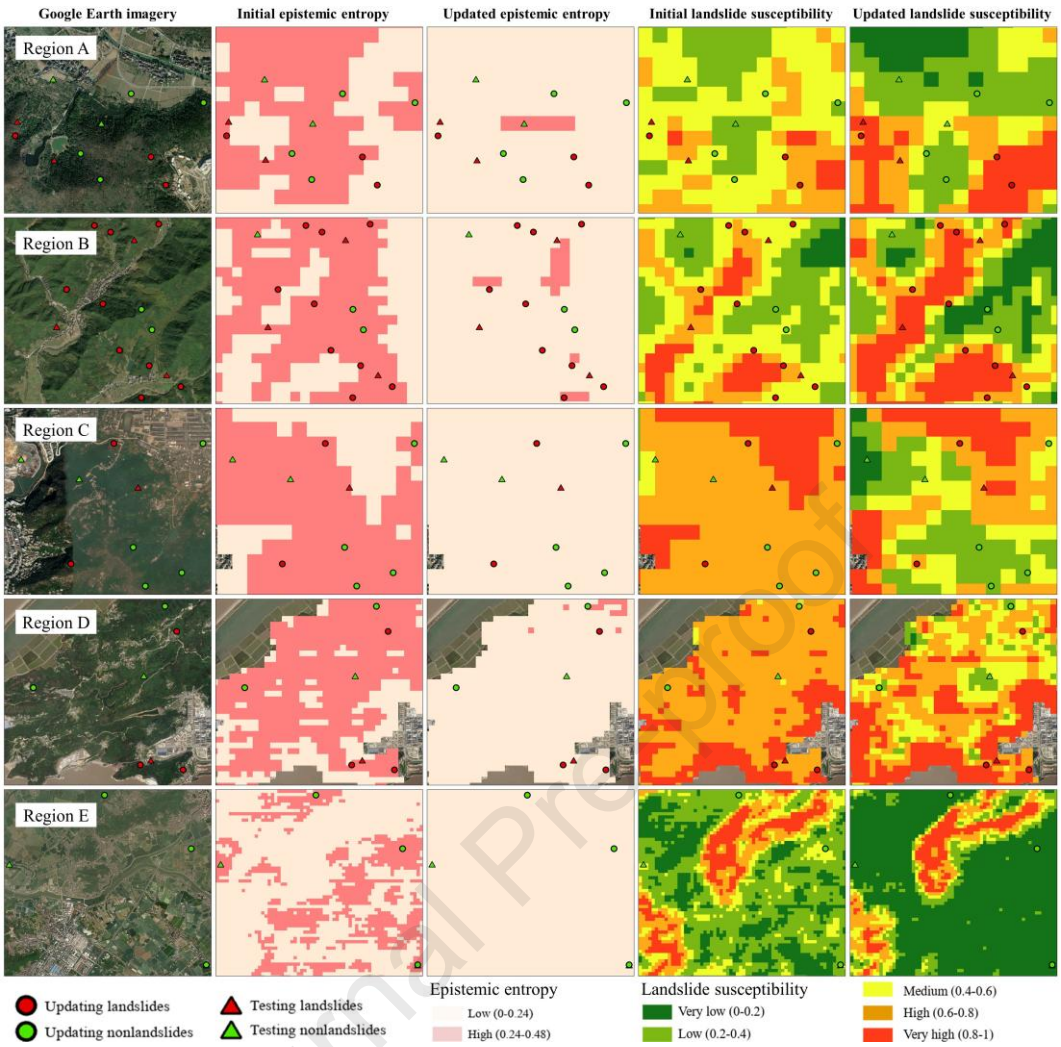

Fig. 12. Zoomed-in views of initial and updated landslide susceptibility maps in high epistemic entropy reduction areas.

5.2. Interpretation of landslide susceptibility in Zhejiang Province

Understanding the mechanics of LSA provides critical insights and support for landslide prevention. In this study, we conducted sensitivity analyses to assess the relative importance of 16 LCFs across Zhejiang Province. The importance of each LCF was quantified using the AUC reduction, comparing scenarios where each LCF was replaced by its spatially invariant mean value while allowing all other factors to vary (Fig. 13). Our findings indicated that slope emerged as the most influential LCF, followed by NDVI. Specifically, when slope and NDVI were held globally constant, the AUC reductions were 0.068 and 0.049, respectively—significantly higher impacts compared to other LCFs. Spatially uniform values for the remaining LCFs led to only minor performance degradations, with only roughness above 0.01. Consistent with our study, Wang et al. (2020) found that landslide susceptibility in Zhejiang Province was dominated by roughness, NDVI, and slope, although annual precipitation was also important at the 180m pixel scale. Compared to more urbanized regions, such as Hong Kong, China (Chen et al., 2023), NDVI plays a more important role for LSA in Zhejiang Province, highlighting the role of vegetation in regulating landslide risk in vegetation-rich environments. Incorporating seasonal NDVI into the LSA model can enhance assessment accuracy by better accounting for variations in vegetation cover (Chen et al., 2023). However, the cloudy summer conditions in the study area limited NDVI acquisition, resulting in the neglect of seasonal variations in vegetation cover, which warrants further investigation. Notably, the geological LCFs (DTF, lithology) exhibit consistent insignificance, likely due to the predominance of shallow landslides which are less influenced by deep-seated geological structures.

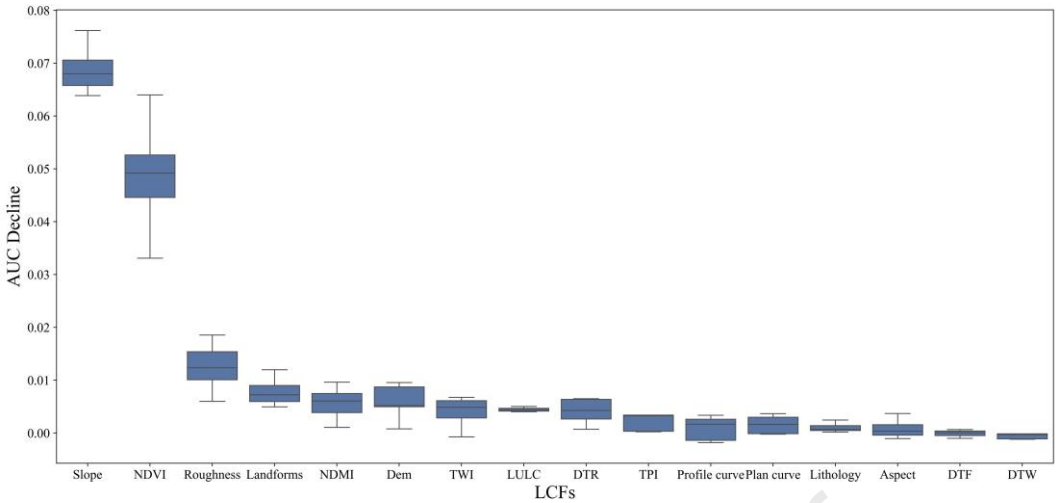

Fig. 13. Feature permutation importance of U-BGWCNN in Zhejiang Province.

The geographically weighted attention module employed provided deeper insights by assigning varying local weights to each LCF. By integrating the distance matrix from each geographic reference point into our model, we visualized the spatial distribution of geographically weighted attention for the top four LCFs (Fig. 14). Our analysis revealed that slope is an especially key predictor on eastern hills, northern plains, and the intersection of the central basin and southern mountains. Slope not only plays an important limiting role in plains but also assumes importance in terrain intersections or mountainous areas where gradients can vary significantly. NDVI exhibits higher attention toward the southern mountains and northwestern hills, however, lower attention to the plains. The rich vegetation cover in mountainous areas and extensive cultivation of economic crops have led to significant NDVI variations in Zhejiang (Chen et al., 2023). The higher importance of NDVI in predicting landslide risk in mountain regions may be due to the significant relationship between vegetation cover and landslide occurrence. Conversely, smaller NDVI differences in plains, coupled with negligible landslide risk, make NDVI less important for LSA in these regions. Roughness is more important for the Zhoushan Islands LSA, likely due to its complex coastlines and rugged terrains with varying elevations, resulting in a strong relationship between roughness and landslide risk. Landform pays more attention to complex mountainous areas in the northwest, where comprehensive topographic information aids in LSA, particularly in areas with intricate terrain. By analyzing the spatial variability attention of these LCFs, we can provide targeted guidance for developing landslide mitigation strategies in different regions. For instance, enhancing vegetation cover in mountainous areas effectively reduces landslide risk, while adopting landform-specific risk prevention measures is critical for managing landslide risk in areas with complex topography.

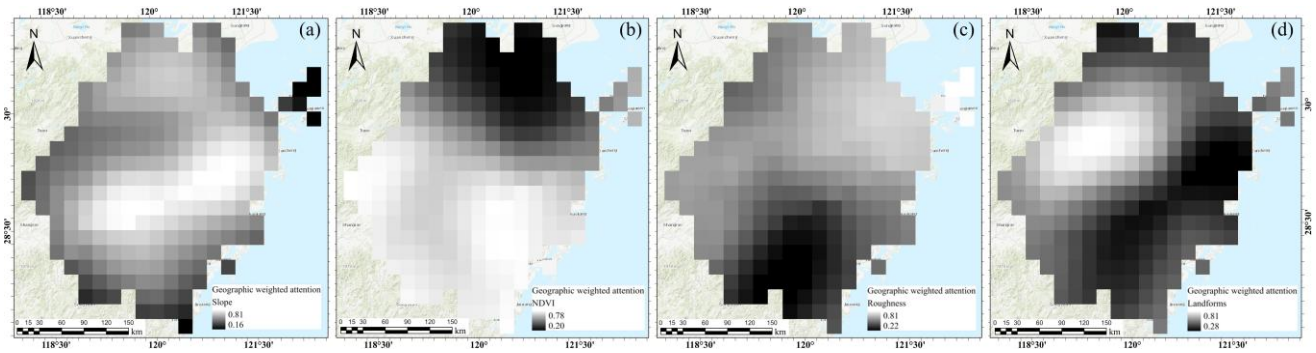

Fig. 14. Geographically weighted attention of the main LCFs: (a) Slope, (b) NDVI, (c) Roughness, and (d) Landforms.

5.3. Flexibility and generalizability of the framework

The advantages of the framework lie in its flexibility and wide applicability. Various DL models, such as GNN (Lin et al., 2025), RNN (Caceres et al., 2021) and Transformers (Xue et al., 2021) can be selected as the functional model for BDL to meet the specific needs of LSA in different study areas or research priorities. For instance, this study utilized BGWCNN to effectively capture the spatial heterogeneity of landslide risk in Zhejiang Province,

characterized by its complex topography (Fig. 1). If high model transferability is desired, employing a global DL model may be more suitable as the functional model. Additionally, modifying the input structure of the functional model by RNN helps accommodate multi-year rainfall data for LSA (Gao et al., 2024). Since BDL models are inherently data-driven, the framework applies to LSA across diverse climatic conditions and triggering mechanisms, including seismic activity (Shao et al., 2020; Yang et al., 2024), and glacial melt (Zhou et al., 2024; Chang et al., 2025). Similarly, using InSAR to monitor and identify landslides is not limited to the study area but is equally applicable to other regions, such as arid areas (Liu et al., 2021a; Jiang et al., 2025) and regions impacted by glacial melting (Kos et al., 2016; Kim et al., 2022). By leveraging its flexibility and generalizability, the framework can dynamically identify epistemic deficits across various LSA tasks (e.g. changes in soil moisture due to glacier retreat, terrain and fault alterations due to earthquakes) for ongoing updates.

5.4. Limitations and future work

Rainfall, a triggering factor for most landslides in Zhejiang Province, was excluded from this study. Meanwhile, we ignored the uncertainty of time intervals in LSA, yet rainfall data across different time intervals can exhibit vastly different distributions, leading to different mechanisms of landslide occurrence (Chen et al., 2023; Zhao et al., 2024). While incorporating long-term rainfall data (multi-year average) into LSA models can provide insights into dynamic LSA, the inherent variability and randomness of rainfall events often limit the effectiveness of these models for precise landslide predictions (Jones et al., 2021). To enhance the integration of rainfall into LSA modeling, combining spatial probabilities of LSA with the temporal probabilities of rainfall events (e.g. intensity-duration curves (Ma et al., 2015) and data-driven models (Mondini et al., 2023)) could serve as a more effective strategy for dynamic LSA (Jiang et al., 2024). Although C-band Sentinel-1 InSAR is prevalent for its accessibility and broad coverage, its effectiveness in areas with dense forests and steep terrain can be reduced (Fig. 6). Utilizing both ascending and descending orbit InSAR data by combining datasets from different look angles can reduce the impact of shadows and layovers caused by steep topography (Jiang et al., 2025). L-band SAR products, known for their reduced sensitivity to vegetation, enhance coverage in areas with dense vegetation (Liu et al., 2021b; Mondini et al., 2021). Additionally, integrating multi-source data from GNSS provides more accurate detection of terrain deformation in key areas (Qin et al., 2022; Huang et al., 2023). By employing multi-angle, multi-band, and multi-source InSAR, monitoring landslide deformation can be more extensive and accurate. Furthermore, the BDL performance is heavily influenced by hyperparameter choices, including priors, initialization settings, and loss function weight assignments, all of which require substantial research expertise. Optimizing the BDL hyperparameters through heuristic algorithms can reduce computational requirements while ensuring robust and accurate model performance.

6. Conclusions

This study introduced a new BDL framework to update LSA through epistemic uncertainty, using augmented samples in high epistemic entropy areas by E-SBAS-InSAR; we achieved a full-regional time-updated LSA in Zhejiang Province, China. Based on the findings, several conclusions can be drawn:

(1) The BGWCNN model demonstrated significant improvements over traditional ML models, achieving a 3.2% increase in AUC compared to the state-of-the-art RF model while maintaining the best stability.

(2) Updating based on epistemic entropy fills knowledge gaps and improves model performance. The high epistemic entropy region of the U-BGWCNN accounts for only 0.2% of the overall area, representing a 90.9% reduction compared to the initial epistemic entropy region. U-BGWCNN further improved the BGWCNN with OA of 2.7%, F1-score of 2.3%, and AUC of 2.5%.

(3) Slope emerged as the most influential factor for LSA in Zhejiang Province, followed by NDVI. Slope is more concerned with multiple terrains and their intersections, whereas NDVI plays a greater role in mountainous areas than in plains.

CRediT authorship contribution statement

Chongzhi Chen: Conceptualization, Methodology, Software, Validation, Data curation, Formal analysis, Investigation, Writing - original draft, review & editing, Visualization. **Baiyu Dong:** Conceptualization, Methodology, Writing - original draft, review & editing. **Jingya Lin:** Visualization. **Zhangquan Shen:** Methodology. **Yuhui Weng:** Investigation, Writing - review & editing. **Luming Fang:** Methodology, Data curation. **Ke Wang:**

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Conceptualization, Resources, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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